

System Design (Telusko) - Day 2

One-Shot 5h course | Video 0:15-0:30

Scalability - Vertical vs Horizontal

How systems grow to handle more load: scaling up one machine vs scaling out to many.

Key concepts

- Vertical scaling: bigger machine (more CPU/RAM). Simple but has a ceiling and a single point of failure.
- Horizontal scaling: add more machines behind a load balancer. Scales far but needs statelessness.
- Stateless services scale horizontally easily; push state to a DB/cache.
- Trade-offs: cost, complexity, fault tolerance.

Key example

Rule of thumb: design services to be stateless so any instance can serve any request.

Today's tasks (1.5h)

- [] Compare vertical vs horizontal for a web API
- [] Identify what state blocks horizontal scaling
- [] Sketch a scaled-out architecture

Resources

Video: youtu.be/Vnm-ycSfJx4 (start at 0:15-0:30)

Docs: docs.telusko.com/docs/system-design

(Full clickable links are in the app: Goals > this day > Study block.)